

Single-trial cortical prediction of the H-reflex during operant conditioning

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Summary. Operant conditioning of the spinal H-reflex produces lasting change in the spinal cord, and sensorimotor cortex is known to influence the reflex pathway (Boulay et al., 2015). Whether cortical activity *changes* as the animal learns has not been established. We decoded 2.7 million trials of paired soleus EMG and sensorimotor ECoG from six chronically implanted rats (three down-conditioned, three up-conditioned) recorded on the Elizan III system, and trained a self-supervised autoencoder on the cortical waveform. Three results follow. First, conditioning reproduced in both directions after controlling for stimulus intensity and motoneuron-pool excitability. Second, the naive measure — drift of the average cortical state across conditioning — does not survive its own controls and reflects slow recording nonstationarity rather than learning. Third, using a design in which decoders are trained and tested *within* five-day blocks, cortical activity predicts single-trial H-reflex amplitude above a within-block shuffle null in 99 of 101 blocks across all five animals (animal-level $t(4) = 3.81$, $p = 0.019$), contributing roughly 2% of reflex variance uniquely once stimulus and background are accounted for. The effect is small in magnitude but near-perfectly reliable. Across animals, the change in this coupling correlates with how much each animal learned ($r = +0.64$, $n = 5$), though the sample is not yet sufficient for significance.

1. Background

In the operant conditioning paradigm developed in this laboratory, rats are rewarded for increasing or decreasing the size of the soleus H-reflex, the largely monosynaptic electrical analogue of the spinal stretch reflex. Over several weeks the reflex changes, and the change persists in the spinal cord itself; a shift of 20% or more is the laboratory's criterion for successful conditioning (Thompson and Wolpaw, 2014). Boulay, Chen and Wolpaw (2015) showed that ongoing sensorimotor cortex activity influences H-reflex size in awake behaving rats, establishing that a cortical–spinal relationship exists. The question we address is whether that relationship itself is altered by learning, and whether it is detectable on individual trials.

2. Data and methods

2.1 Dataset and decoding

Recordings were made on the Elizan III system between 2005 and 2007 and archived as MyISAM tables. Each stimulus produces one trial containing a 30 ms window sampled at 5 kHz on two channels: soleus EMG and sensorimotor ECoG. We reconstructed the binary format directly; contrary to the archived decoder documentation, samples are stored *little-endian* as 16-bit integers in block channel order, with a conversion factor of 2.441406 μV per unit. Decoding as documented yields noise. Conditioning onset and direction were recovered per animal from the experimenters' typed annotations. Six animals passed quality control, comprising 2.7 million trials over 45 to 210 days each.

2.2 Reflex measurement

Raw H-reflex amplitude is not interpretable in isolation because it scales with stimulus intensity; in these recordings the M-wave drifted from roughly 100 to 450 μV over the course of an experiment,

which alone can reverse the apparent direction of an effect. We therefore measure, per trial and after background subtraction, the M-wave (2–4 ms) as a receipt for delivered stimulus, the H-reflex (6–9 ms), and the pre-stimulus background EMG in the 20 ms preceding the stimulus as an index of motoneuron-pool excitability. Both the M-wave and the background are removed from every subsequent measure by regression, including quadratic terms.

2.3 Cortical representation

A one-dimensional convolutional autoencoder compresses each 150-sample cortical window to a 48-dimensional code, trained on 600,000 trials pooled across five animals with per-animal standardisation and no access to any label. The encoder is then frozen and used purely as a feature extractor.

2.4 Controls

Four controls constrain every cortical claim. A *baseline sham* splits pre-conditioning data as though it were an experiment; nothing occurred, so an honest measure must report approximately zero. A *randomised-order* null shuffles the assignment of trials to days, preserving the numbers while destroying time structure. A *direction contrast* exploits the fact that electrode drift is independent of whether an animal was trained up or down, whereas learning is not. Finally, coupling analyses are cross-validated *within* five-day blocks, so slow drift displaces training and test trials together and cancels.

3. Results

3.1 Conditioning reproduces in both directions

After correction for stimulus and background, corrected H-reflex amplitude rises in up-conditioned animals and falls in down-conditioned animals. Three of five analysed animals meet the 20% criterion in the trained direction; the two that do not serve as internal controls below. Change is expressed as a percentage of the baseline level, which carries a caveat: one animal (A3) has a baseline H of 19.4 μV against 60–167 μV in the others, so its percentage change (+249%) overstates the underlying change relative to the other animals. The direction of change is unaffected. A symmetric measure ($\log_2$ fold change of the covariate-matched reflex) is planned for the full sample.

3.2 Average-state drift does not reflect learning

The intuitive analysis — measuring how far the mean cortical representation moves from baseline — yields a large apparent effect that does not survive scrutiny. The baseline sham drifts as much as genuine conditioning in five of six animals; the drift is largest at conditioning onset and declines during the period of strongest behavioural change; and there is no dependence on conditioning direction (up 21.9 vs down 34.6, $t = -1.01$). The randomised-order null is approximately zero throughout, indicating the metric is not an artefact of the data values themselves. We conclude that this drift reflects slow change in the recording rather than learning. Within a single animal, learning and electrode drift are both smooth functions of time and are therefore not separable; this is a limitation of the design, not of the data.

3.3 Cortical activity predicts single-trial reflex amplitude

Reframing the question removes the confound. Decoders trained and tested within five-day blocks predict single-trial H-reflex amplitude above a within-block label shuffle in 99 of 101 blocks and in all five animals (mean cross-validated $R^2 = 0.037$, null = -0.009 ; treating animal as the unit of analysis, $t(4) = 3.81$, $p = 0.019$; the block-level sign test is not quoted because blocks within an animal are not independent). Judged by magnitude the effect is small; judged by reliability it is close to invariant. Removing stimulus and background *within* each block rather than globally is essential: the global procedure leaves between-block stimulus variance, which cortex tracks, and inflates the estimate to 0.090. We report the conservative value.

3.4 Variance partition

Partitioning variance in H-reflex amplitude clarifies the magnitude of the cortical contribution.

Source	Variance explained	Interpretation
Stimulus (M-wave)	47%	Dominant factor
Background excitability	1%	Small but a necessary control
Cortical activity, alone	21%	Largely shared with stimulus
Cortex, unique	2%	Added once the others are known
Unexplained	51%	Residual trial-to-trial variability

Cross-validated within blocks, averaged over five animals. Cortex and stimulus share substantial variance because the stimulus drives both the cortical evoked response and the reflex; only the non-overlapping portion is counted as unique.

3.5 The effect is not muscle contamination

Because both channels are recorded simultaneously, the ECoG electrode could in principle register muscle activity. Correlating cortical against muscle band power across trials gives $r = +0.057$ above 100 Hz, the range in which muscle artefact would appear, with slightly negative correlations in the lower bands. A lead-lag comparison places cortical power marginally ahead of muscle power. Contamination is therefore an unlikely explanation.

3.6 The effect scales with learning

Animals differ in how much they learned, and two failed to reach criterion in the trained direction. Across animals, the change in cortical coupling from baseline to conditioning correlates with behavioural change ($r = +0.64$, $t = 1.44$, $n = 5$). The two successful animals for which coupling could be compared gained coupling ($+0.023$ and $+0.022$); the two unsuccessful animals did not (-0.012 and $+0.000$). The direction is that predicted if the coupling is tied to conditioning, but the

sample is too small for significance.

4. Discussion

Two conclusions are supported. Operant conditioning of the H-reflex is reproducible in both directions under proper stimulus control, and cortical activity carries a small, consistent, single-trial signal about reflex amplitude that is not attributable to stimulus intensity, motoneuron-pool excitability, recording drift, or muscle crosstalk. This extends Boulay et al. (2015) from a population relationship to a single-trial one, and situates its magnitude relative to the other contributors.

A third conclusion is methodological and, we suggest, worth reporting. The obvious analysis — tracking movement of the mean cortical state across weeks — produces a convincing-looking result that three independent controls reject. Investigators applying representation-learning methods to long chronic recordings should expect this confound and test for it directly.

Limitations

The sample is five analysed animals, of which two are up-conditioned; a survey of the archived reward criteria indicates no further successfully up-conditioned animals with usable cortical recordings exist in this dataset, so the direction contrast cannot be strengthened. One animal (A12) was excluded because its pre-stimulus cortical signal was 2.8 μV against 106–128 μV in the others, indicating electrode failure. Whether coupling strengthens with conditioning is not resolved at this sample size; the remaining down-conditioned animals would power that test.

Acknowledgements

We thank Dr. Jonathan S. Carp and Theresa M. Vaughan of NCAN for guidance throughout, and in particular for proposing the baseline sham control, the randomised-order control, the pre-stimulus excitability covariate, and the scaling analysis, each of which materially changed the conclusions. This work uses recordings collected in the Laboratory of Neural Injury and Repair at the Wadsworth Center.

References

- Boulay CB, Chen XY, Wolpaw JR. Electrographic activity over sensorimotor cortex and motor function in awake behaving rats. *Journal of Neurophysiology* 113: 2232–2241, 2015. doi:10.1152/jn.00677.2014
- Thompson AK, Wolpaw JR. Operant conditioning of spinal reflexes: from basic science to clinical therapy. *Frontiers in Integrative Neuroscience* 8: 25, 2014. doi:10.3389/fnint.2014.00025

Reproducibility. All decoding, analysis and figure code, the per-animal numerical results, and the presentation materials are available at github.com/switchrao777/hroc-ncan. Raw recordings remain with NCAN.

Figures

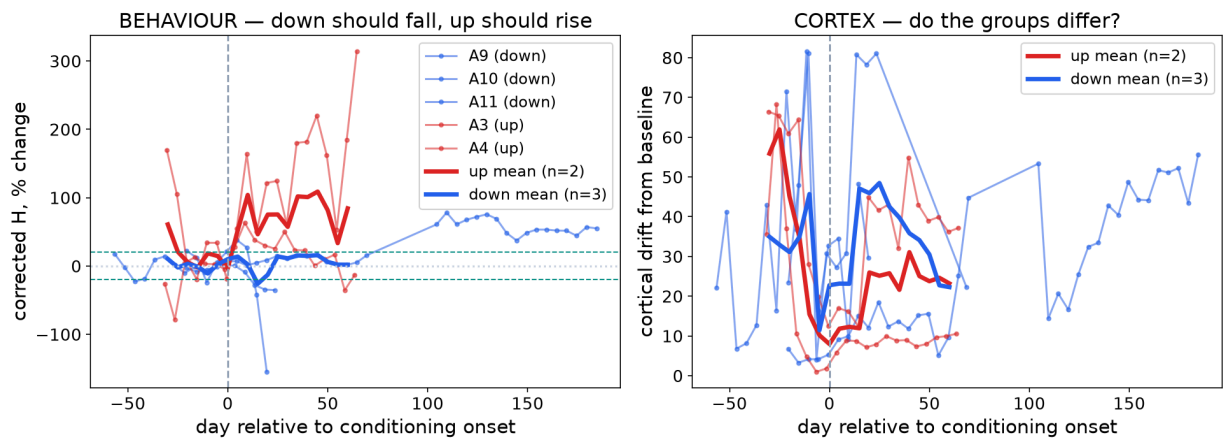


Figure 1. Behaviour and cortical drift, each animal aligned to its own conditioning onset. Left: corrected H-reflex amplitude rises in up-conditioned animals (red) and falls in down-conditioned animals (blue). Right: cortical drift shows no dependence on conditioning direction.

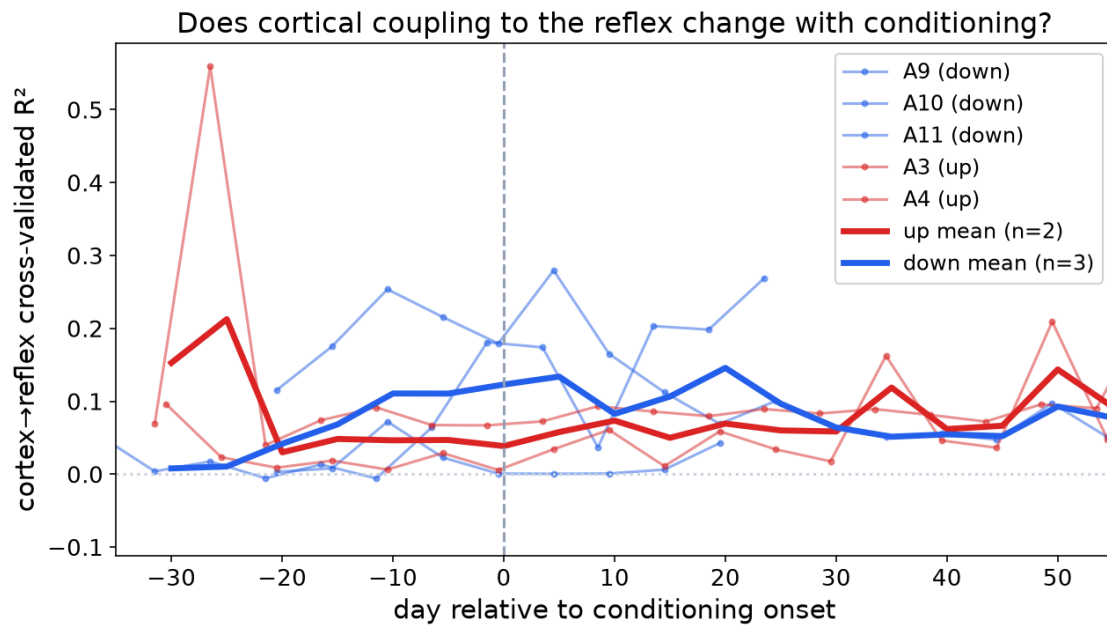


Figure 2. Cortex-to-reflex coupling over time. Cross-validated R^2 within five-day blocks, after removal of stimulus and background. Coupling exceeds the within-block shuffle null in every animal.

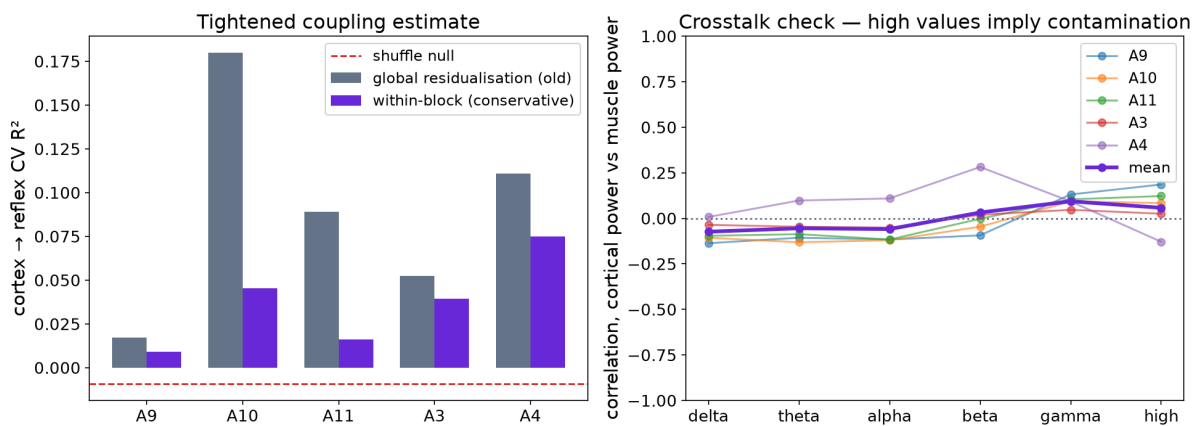


Figure 3. Left: the conservative within-block estimate against the earlier global procedure. Right: correlation between cortical and muscle band power; values near zero above 100 Hz argue against muscle contamination.

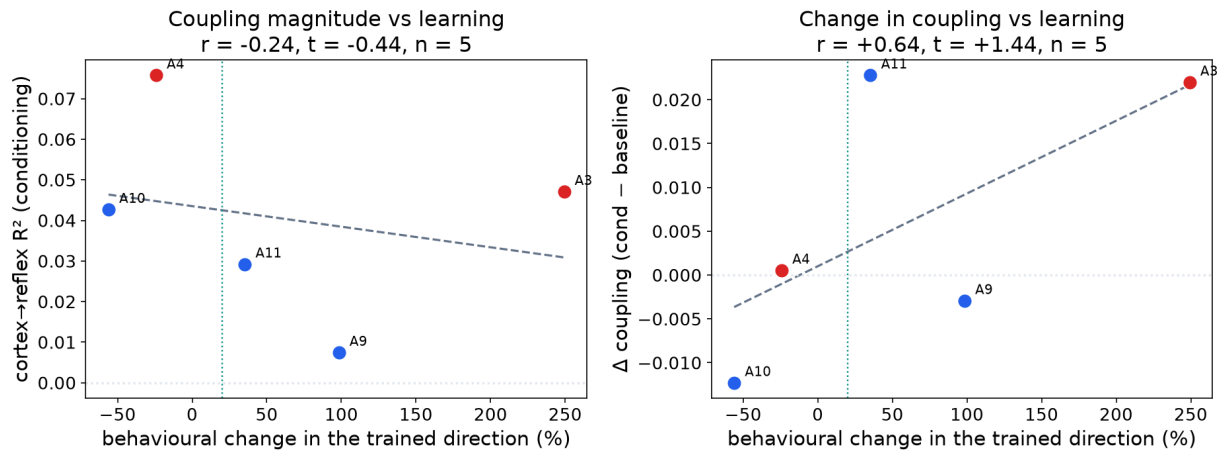


Figure 4. Cortical coupling against behavioural change. Animals that learned more show a larger increase in coupling ($r = +0.64$); the two animals that failed to reach criterion do not.